ROOT LOCUS: DESIGNING A GAIN FOR A DAMPING TARGET

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Sketch the root locus of the proportional angle control loop using the construction rules, and check it in MATLAB.	☐
2	Use the root locus to choose a proportional gain that meets an overshoot requirement, and predict the settling time.	☐
3	Verify the design on the DC motor, and explain from the locus why proportional control alone cannot make the response settle faster.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The angle of the shaft is measured from the encoder counts, as described in the practical exercise, Manual PID Tuning of DC Motor Angle Control, and the output voltage of the controller sets both the direction and the PWM value. Because the angle is the integral of the speed, the angle model is $G_\theta(s) = K_\theta/(s(\tau s + 1))$, where $K_\theta = 6K$ converts RPM/V into degrees per second per volt. For the A1.4 example motor, $K_\theta = 168$ °/(s·V).

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. The root locus of the angle loop shows how the closed-loop poles move as the proportional gain increases. It is used to choose the gain that gives an overshoot below 5%, and the design is then tested on the hardware. The exercise supports the ACE-CORE learning guide, Proportional Control Design using Root Locus.

Sketch the Root Locus

The open-loop transfer function $K_p K_\theta/(s(\tau s + 1))$ has two poles, at $s = 0$ and $s = -1/\tau$, and no zeros. The construction rules give:

- The locus lies on the real axis between the two poles.
- The two branches meet at the breakaway point, halfway between the poles, at $s = -1/(2\tau)$.
- With two more poles than zeros, the branches leave along two asymptotes at ±90°, centred at the breakaway point.

So, once the gain passes the breakaway value, the closed-loop poles move straight up and down a vertical line at $s = -1/(2\tau)$. The requirement of an overshoot below 5% corresponds to $\zeta \geq 0.7$, a line from the origin at $\cos^{-1}(0.7) \approx 45.6°$ to the negative real axis. The design gain is where the locus crosses this line:

$$K_p = \frac{1}{4\zeta^2 K_\theta \tau}, \qquad s = -\frac{1}{2\tau} \pm j\frac{\sqrt{1-\zeta^2}}{2\zeta\tau}, \qquad t_s \approx \frac{4}{\zeta\omega_n} = 8\tau$$

The root locus for the example motor is shown in Figure 1. The design gain places the poles at s = -3.33 ± j3.40, giving a predicted overshoot of 4.6% and a settling time of about 1.2 s.

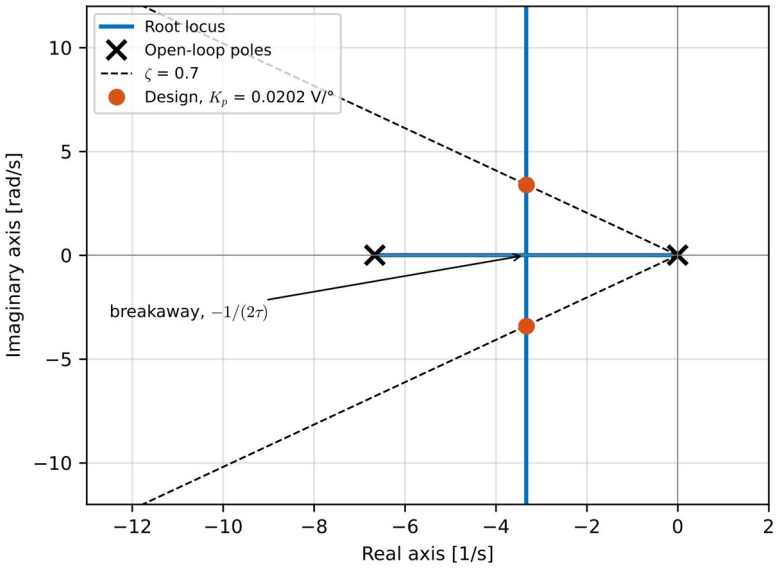


Figure 1: *Root Locus of the Angle Loop for the A1.4 Example Motor, Showing the ζ = 0.7 Design Point (K_p = 0.0202 V/°)*

1. Sketch the root locus for your motor by hand, marking the poles, the breakaway point and the asymptotes.
2. Draw the $\zeta = 0.7$ line and find the design gain graphically and from the equation above.
3. Check your sketch with `rlocus(tf(Ktheta,[tau 1 0]))` and `sgrid(0.7,[])` in MATLAB, and click on the locus to read the gain at the crossing.

Verify the Design on the Hardware

1. Use the proportional angle control model from the practical exercise, Second-Order Response: DC Motor Angle under P Control, without the measurement filter.
2. Set K_p to your design gain and apply a 90° step. Log the angle and measure the overshoot and settling time.
3. Repeat with twice and four times the design gain, and record all three results in Table 2.

Table 2: *Results of the Root Locus Design*

Gain	K_p (V/°)	Predicted OS (%)	Simulated OS (%)	Measured OS (%)	Measured t_s (s)
1 × design	0.0202	4.6	6.2		
2 × design	0.0405	16.7	21.2		
4 × design	0.0810	30.9	36.8		

The simulated overshoot in Table 2 is simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares (K = 28 RPM/V, τ = 0.15 s), so they show the behaviour to expect rather than measured data. The simulated overshoot is slightly higher than predicted at every gain, because the sampling and one-sample delay of the digital controller add lag that the continuous-time model ignores; the practical exercise, Root Locus on the z-Plane: Discrete P Design for Angle Control, accounts for this. The simulated settling time barely changes between the three gains (about 1.3 s, 1.3 s and 1.5 s), because the real part of the poles is fixed at $-1/(2\tau)$ once they leave the real axis.

1. Explain, using the locus, why increasing K_p past the breakaway point does not reduce the settling time.
2. What kind of compensator would bend the locus to the left, and how would that help?
3. How would the locus change with the measurement filter from the practical exercise, Routh-Hurwitz: Finding the Critical Gain, added?

4.0 Concluding Remarks

This practical exercise used the root locus to design a proportional gain for the angle of a DC motor shaft. Using MATLAB, Simulink and ACE-Lab hardware, the locus was sketched from its construction rules, the gain meeting a 5% overshoot requirement was read from it, and the design was verified on the motor.

The locus showed not only which gain meets the requirement, but also the limits of the controller: with proportional control alone, the poles cannot move to the left of

$-1/(2\tau)$, so the settling time cannot be improved by raising the gain. Seeing such limits before building a controller is the main strength of root locus design.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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